# Ziyun Lan

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## Education

**Aug 2026 – Present** **National University of Singapore** **Computing (Computer Science) | Master**

**Sep 2022 – Jun 2026** **Nanjing Normal University** **Computer Science and Technology | Bachelor**

- **Cumulative GPA**: 3.9/5, **Ranking**: 5/113
- **Core Courses**: Java Programming, Machine Learning, Algorithm Design, Computer Organization and Architecture, Operating Systems

## Project Experience

**Jul 2026 – Aug 2026** **RAG and Agent-based Intelligent Agent for Robot Vacuum Cleaners** **Personal Project**

- **Project Overview**: Developed a RAG-Agent Q&A system with retrieval & tool-calling for Q&A and report generation.
- **Responsibilities**: **1.** Built ReAct Agent via LangChain, wrapping RAG, weather, location & usage-record tools; **2.** Deployed Chroma for document loading, recursive splitting, embedding & recall; **3.** Implemented context tagging & dynamic prompts for personalized reports using external data; **4.** Created Streamlit UI with logging, error-handling & MD5 deduplication for incremental KB updates.

**Jul 2026 – Aug 2026** **EyeVQA: Ophthalmic VQA Benchmark for Recognition and Spatial Grounding** **MICCAI Workshop (Oral)**

- **Project Overview**: Constructed an ophthalmic VQA benchmark for disease recognition, cross-image reasoning and spatial grounding.
- **Responsibilities**: **1.** Aggregated 21 ophthalmic datasets; preprocessed images in Python, compiled 27,935 fundus images and constructed 20,000 JSON-formatted QA pairs; **2.** Generated ground truth from diagnostic labels, severity grades, mWasks and anatomical annotations, designed 7 question types; **3.** Conducted zero-shot evaluation of 14 VLMs and computed metrics including accuracy and IoU..

**Sep 2024-Jul 2025** **YOLO-based Intelligent Parasite-Egg Detection System for Microscopic Images** **Nanjing CDC Project**

- **Project Overview**: Developed an automatic pipeline eliminating inefficiency and errors in manual microscopic parasite egg observation.
- **Responsibilities**: **1**. Optimized image stitching via SIFT feature matching and slope filtering to build global coordinate system; **2.** Trained YOLOv8n model with active learning for fast parasite egg detection; **3.** Realized egg deduplication through coordinate transformation and Euclidean distance algorithm; **4.** Integrated interfaces to achieve automatic accurate counting, panorama and coordinate output.

## Internship Experience

**Oct 2025-Mar 2026** **The 14th Research Institute of CETC** **Software Development / Algorithm Intern**

**AFSIM Parsing**: Leveraged Qt for recursive file parsing & UI interaction, regex-extracted config to link platforms, routes and sensors.

**Aircraft Trajectory Classification System**: Built trajectory classification visualization for rule-based prediction limitation. **1.** Generated normalized-image trajectory samples with Python; **2.** Built PyTorch-based CNN with data augmentation; **3.** Designed binary-anomaly + fine-grained multi-class pipeline, visualized outputs with Matplotlib.

**Rag-base Q&A System**: Deployed Ollama, RAGFlow & local LLM on Docker, implemented chunking, vector indexing & QA interface.

**Aug 2025-Oct 2025** **Nanjing Citizen Card Payment Co., Ltd.** **Robotic Arm Development Intern**

**Robotic-Arm Environment Setup**: Documented JetArm hardware & wiring, configured Linux and ROS2, deployed NoMachine and SSH.

**Citizen-Card Sorting & Counting System**: **1.** Researched and adapted lightweight YOLO, tuned parameters to improve card-positioning accuracy; **2.** Used Python to fetch 3D coordinates from depth camera, optimized height logic for precise robotic-arm positioning; **3.** Tuned motion & gripping parameters to realize stable grasping and sorting; **4.** Built counting module for automatic total-quantity statistics.

**Mar 2025-Aug 2025** **Nanjing Borun Brain-Inspired Intelligence Technology Co., Ltd.** **Algorithm Intern**

**Helmet Object-Detection**: Performed Labelme annotation for YOLO, split datasets & configured YAML for auto-training and surveillance.

**Multi-Label Behavior Classification Model**: **1.** Modified YOLO to lift single-label constraints; **2.** Customized network with new activation & loss functions for multi-scenario adaption; **3.** Built ResNet18 multi-label model with Dropout to mitigate overfitting; **4.** Developed API-&-LLM-powered auto-annotation scripts for label refinement; **5.** Used Git for version control & Docker for environment isolation.

## Honors & Awards

- **Tech Contests**: Provincial 3rd, Computer Design Contest | Twice Provincial 3rd, Lanqiao Cup C++ | Provincial 1st/2nd, National C++ Challenge | Provincial 1st, Advanced Office Contest | National 3rd, Computer Skill Contest | Provincial 2nd, Java Programming Summit
- **Math Contests**: Twice Provincial 2nd, Huajiao Cup Math Contest | Provincial 2nd, National Math Ability Challenge
- **Honors**: 1st/2nd-class Student Scholarships (3 times) | Merit Student | 2 software copyrights
- **Certifications**: Software Designer (Intermediate) | MIIT-certified Internet Platform Dev Engineer | NCRE Level-2 | IT Teacher Qualification

## Technical Skills

- **Skills**: **Languages**: Python | C++ | Java | Verilog; **Deep Learning**: TensorFlow | PyTorch | OpenCV | YOLO | CNN | RNN | LSTM | Transformer; **Frontend**: HTML | CSS | JavaScript | React | Vue; **Backend & Frameworks**: Spring Boot | MyBatis | Qt; **Databases**: MySQL | PostgreSQL; **Agents**: LangChain | RAG | Agent | Prompt | Codex | Trae; **Tools & Platforms**: Git | Docker | Linux | Notebook | Nginx | PPT | Word | Excel | Overleaf
- **Language Proficiency**: IELTS 6.5 (6 in each sub-score); GRE 322 (AW 3.5); CET-6;